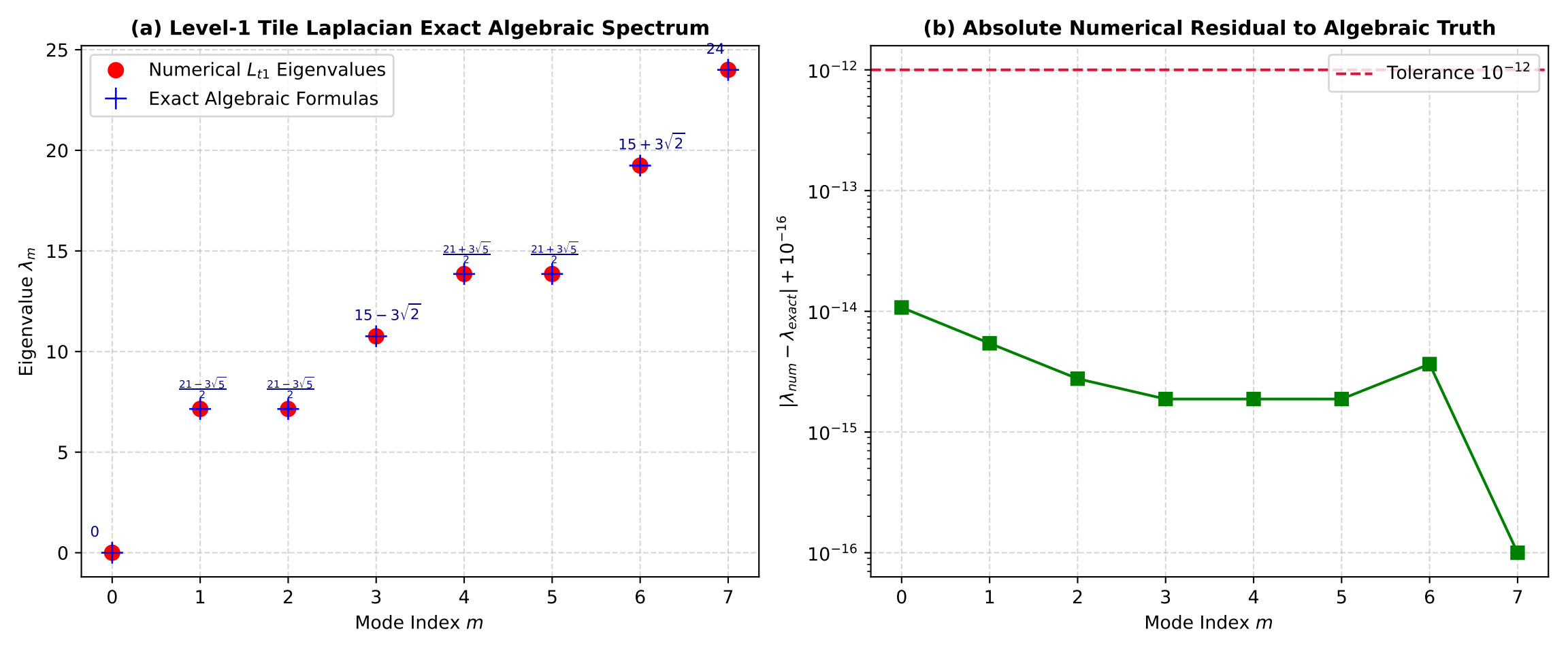
(a) Level-1 Tile Laplacian Exact Algebraic Spectrum
Numerical L_{t1} Eigenvalues
Exact Algebraic Formulas
Eigenvalue λ_m
Mode Index m
0
$\frac{21-3\sqrt{5}}{2}$
$\frac{21-3\sqrt{5}}{2}$
$15-3\sqrt{2}$
$\frac{21+3\sqrt{5}}{2}$
$\frac{21+3\sqrt{5}}{2}$
$15+3\sqrt{2}$
24
(b) Absolute Numerical Residual to Algebraic Truth
Tolerance 10^{-12}
$|\lambda_{num}-\lambda_{exact}|+10^{-16}$
Mode Index m